Self-assessement questions on Strenghtening

- In which circumstances shaping by casting is more beneficial than shaping by strain hardening
- Why impurities in a lattice can induce strengthening in an alloy
- Why the slip systems are preferential planes and directions for dislocation travel
- Why does the interaction beween dislocation and other defects result in a strengthening of the alloy
- Necessary conditions for the formation of a solid solution alloy exhibiting an extensive solute solubility
- Comment on the imortance of Orowan equation (shear strain rate vs. dislocation density)
- What is a common feature in the majority of strengthening methods
- Which region around a dislocation exhibits higher energy/strain in the lattice such that impurities are attracted into
- Explain the manufacturing steps of 7xxx aluminium alloy, from casting to aging heat treatment
- Compare the slip systems in cubic FCC vs BCC
- Explain the consequencies of the interaction between both positive and either positive and negative dislocations
- What is the Peierls-Nabarro stress

Self-assessement questions

- Compare the behaviour of transversal and longitudinal steel texture samples under V-Charphy impact test (CVN)
- How does the CVN of transversal texture compare with the longitudinal one
- Strategies to enhance CVN/ductility in highly textured steel components
- Comment on the simplest test to assess the detrimental presence MnS
- Explain why:
  - a scanning electromagnetic inductor enables the purificarion of an ingot
  - Spraying a liquid metal on rotating metallic disk promote the formation of amorphous ribbons
  - Impurity segregations are not observed during laser surface remelting
- How do you compare the strength and toughness of casting with that of a strain hardened material
- Following the previous question, how strength and toughness can be improved in either processed materials

Self-assessment questions on Lecture n.4

- Compare the slip systems of FCC vs. BCC crystal and their sheet forming ability
- Explain why strengthening by solid solution is proportional to $\sqrt{\%C}$
- Explain how toughness varies after solid solution strengthening of an alloys
- Explain why certain alloy elements cause strengthening in alloys while other elements cause softening
- Explain which lattice defects, other than dislocations, contribute to strain (or work) hardening and why
- Clarify the importance of Orowan's equation for work hardening
- Illustrate the importance of dislocation sources and the conditions under which they operate
- Show the practical implications of Cottrell atmospheres and for which materials and conditions they form
- Clear up when plastic deformation is promoted by twinning rather than by dislocation slip
- Explain the essence of micro- and macro-scopic strain hardening phenomena with reference to both Ludwik-Hollomon and Taylor laws
- Justify the large strain hardening coefficient of Co-alloys and Cu-alloys vs. that of cold rolled steel products
- Delineate which steels exhibit outstanding formablity properties and show inherent reasons
- Elucidate the role of stacking fault energy on dislocation arrangement upon ultrafast plastic deformation
- Compare the consequences of strain aging and dynamic strain aging
- Detail why strengthening by grain size refinement lead to both increased strength and fatigue resistance

Self-assessment questions

- Explain why Al-Si alloy castings are relatively brittle and how toughness can be enhanced
- Discern the influence of precipitates at grain boundaries and those inside the grains on mechanical properties
- Show the relevant factors in precipitation strengthening leading to optimum strengthening after aging
- Provide a sound interpretation of the Orowan equation on precipitation hardening
- Present the distinguishing factors between precipitation hardening and dispersion hardening
- Comment on the microstructure and mechanical properties changes upon the production of Al-Cu alloys